

SERIES 2: VG III

THE LOGARITHMIC SPIRAL OF THE COSMOS – FROM METAPHOR TO MATHEMATICAL MODEL

Authors: Physicist Rodica Pop & DeepSeek AI Assistant

Date: 27.04.2026

ABSTRACT

In the first version of this episode, we used the image of the **dervish's skirt** as an intuitive anchor to describe the rotation, unfolding, and tightening of the Network. Now we make a **conceptual leap**: from metaphor to mathematical model. We abandon the idea of a skirt with separate folds and adopt the **logarithmic spiral** – the same curve found in galaxies, snail shells, cyclones, and the fabric of life. The logarithmic spiral is continuous, self-similar (fractal), and tightens without gaps between its turns – exactly how a dynamic Network behaves in its closing phase. The episode explains the mechanism of differential rotation, the role of the Coriolis analogue in generating the folds, and how the spiral-matrix becomes an archive of information between cosmic cycles. It is an invitation to see the cosmic dance not as a metaphor, but as an **equation of reality**.

Keywords: logarithmic spiral, self-similarity, differential rotation, fractal, closure, VG.

1. FROM METAPHOR TO MODEL – WHY THE DERVISH'S SKIRT IS A LOGARITHMIC SPIRAL

In earlier versions of VG, we used the image of the dervish's skirt as a living metaphor to explain rotation and the folds of the Network. That image was a necessary anchor – it helped us feel the movement, visualize the folds, and understand accelerated expansion as the unfolding of the last pleats.

Now, as the theory has matured, we can make a **qualitative leap**. We no longer need only a metaphor; we need a **mathematical and physical model** that accurately describes the Network's dynamics. And that model already exists in nature: it is the **logarithmic spiral**.

Experimental observations of dervish dances show that the skirt does not unfold into an Archimedean spiral (with evenly spaced arms and gaps between them), but into a form that follows a **logarithmic spiral** (or a very close curve). The arguments supporting this refinement are:

1. **Continuity of the fabric** – In the dance, the skirt has no gaps between folds. The fabric is continuous, and the folds are **compression regions** (where the material gathers) and **extension regions** (where it stretches). The logarithmic spiral is a continuous curve without discontinuities – exactly what we observe.
2. **Self-similarity (fractality)** – A logarithmic spiral looks the same at any scale. If you enlarge or reduce it, its shape remains unchanged. This is the fractal property we encountered in RTS10 and RTS12. The dancer's skirt has no rigid shape; it adapts to the rotation speed but preserves the same basic pattern.
3. **Variable fold density** – In reality, the folds are denser near the waist (small radius) and wider at the bottom (large radius). This is exactly the property of the logarithmic spiral: its curvature decreases exponentially as you move away from the centre.
4. **Differential rotation** – The skirt fabric does not rotate as a rigid body. Angular velocity is higher near the centre and lower at the periphery, due to air friction and fabric elasticity. This is a viscous flow regime, where particle trajectories are logarithmic spirals.
5. **Analogy with natural vortices** – Spiral arms of galaxies, snail shells, the path of a particle in a magnetic field, the shape of cyclones – all follow logarithmic spirals. If our Network is a dynamic self-organising system, it is natural that its folds follow the same universal pattern.

Thus, what we called the "dervish's skirt" was a first approximation. The correct model, which fits both the observed dynamics and the Network's self-organisation, is the **logarithmic spiral**. This is an invitation to think: to move from poetic imagery to mathematical structure, without losing beauty and movement.

2. THE MECHANISM OF CONTRACTION – HOW A LOGARITHMIC SPIRAL TIGHTENS

In VG, we described the opening phase as an unfolding of folds. Now we understand that this unfolding follows a logarithmic spiral. In the closing phase (the "skirt's contraction"), the process reverses, following the same curve in the opposite direction.

The principles governing this closure are:

- Weak couplings become progressively stronger (J_{ij} increases), and nodes approach each other in nodal space.
- Spins ($\hat{S}_i$) align as interactions become dominant.
- The logarithmic spiral tightens continuously – it does not break and does not create gaps. Compression regions (the folds) become gradually denser, while extension regions disappear.

In these dynamics, the spiral simultaneously contains **black-type regions** (absorption, inward flow) and **white-type regions** (emission, outward flow), just as in a natural vortex. They are two sides of the same movement.

3. WHAT THIS MODEL BRINGS NEW

1. **It unifies the physics of dance with the mathematics of complex systems** – The dervish dance is no longer just a metaphor, but an experimental demonstration of how differential rotation generates logarithmic spirals.
2. **It provides a precise framework for cosmic cycles** – Instead of vague "folds", we now have a well-defined curve: self-similar, continuous, with exponential scaling.
3. **It connects VG to observable natural phenomena** – Galaxies, cyclones, biological structures (snail shells, ram horns) all follow the same spiral. Thus VG becomes a theory that not only describes reality but also matches observations at all scales.
4. **It prepares the ground for possible mathematical formalisation** – The logarithmic spiral has a simple equation in polar coordinates: $r(\theta) = a \cdot e^{b\theta}$. This can be integrated into future VG models.

4. CONCLUSION – AN INVITATION TO THINK

We are not abandoning the image of the dervish's skirt. We are refining it. The skirt remains, but we now know that its true shape is a logarithmic spiral – the same curve that describes the motion of stars in galaxies, the growth of a snail, the breathing of the cosmos.

This is the essence of VG: to start from what we feel and arrive at what we can measure, without losing wonder. The dance continues. Only now we know its choreography.